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# Computer Applications for Popular Music Composers in Creative Work and Teaching

## Introduction

For more than a decade, technological progress and the associated digitalisation of virtually all spheres of life has been an accelerating and tumultuous process. The community of people using information technology in various aspects of their everyday lives has been dubbed the information society[1]. Nowadays, this population uses not only computers, but various computerised devices, allowing it to download and upload a great amount of multimedia content. This content consists of images, sound, and text in diverse inter-relations, together constituting a virtual world[2]. The ever-expanding market accessibility

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1 F. Webster, *Theories of the Information Society*, Third edition, Routledge, London/New York 2006, pp. 263–273; H.K. Nath, *The Information Society*, "Journal of the Singapore Chinese Teachers' Union" 2009, vol. 4, 2009, pp. 19–29; E. Ziemba, T. Papaj, R. Żelazny, *New perspectives on information society. The Maturity of Research on a Sustainable Information Society*, "Online Journal of Applied Knowledge Management" 2013, vol. 1, issue 1, p. 52.

2 G.A. Tsihrintzis, L.C. Jain, *Multimedia Service in Intelligent Environments. An Instroduction*, in: G.A. Tsihrintzis, L.C. Jain (ed.), *Multimedia Services in Intelligent Environments,* Springer-Verlag, Berlin/Heidelberg 2008, pp. 1, 2; J.R. Pardo, *Copyright and Multimedia*, Kluwer Law International, The Hague/London/New York 2003, pp. 5, 6; S.M. Rahman, *Multimedia Technologies. Concepts, Methodologies, Tools, and*

of computers, and their broad improvement, along with different multimedia technologies, make an impact on music-related work, which includes composition, musical arrangement and the application of digital technology in teaching at academic institutions[3].

Technological development impacts the work of a musician, since the modern composer is more and more commonly the sound engineer and the producer, which requires the use of a computer. Their expertise is thus much wider than conventional music education, extending into the fields of computer science, technology, sound engineering and recording. Composers involved in electroacoustic music, or electronic music imitating acoustic sound, commonly utilise the MIDI interface, which is a system allowing for connections between electronic devices, including computers and digital instruments. The system facilitates sending data between devices and simultaneous control of each device[4]. MIDI does not determine the quality of the generated sound. Instead, it allows for an exchange of information (such as notes on a virtual score) between different devices[5]. The use of virtual instruments which imitate traditional ones, or allow for new timbres[6] for the sound design of original

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*Applications*, in: S.M. Rahman (ed.), *Preface. What is Multimedia?*, Information Science Reference, Hershey/New York 2008, p. XXVIII.

3 P. Lévy, *Cyberculture*, University of Minnesota Press, Minneapolis/London 2001, p. 122, 125.

4 D.M. Huber, *The Midi Manual. A Practical Guide to MIDI in the Project Studio*, Third edition, Focal Press, Amsterdam 2007, pp. 1–13; J. Rothstein, *MIDI a Comprehensive Introduction*, Second edition, A-R Editions, Madison 1995, pp. 1–14; P.L. Alexander, *How MIDI works*, Sixth edition, Hal Leonard, Milwaukee 2001, pp. 143–148.

5 B. Enders, *From Idiophone to Touchpad. The Technological Development to the Virtual Musical Instrument*, in: T. Bovermann, A. de Campo, H. Egermann, S.I. Hardjowirogo, S. Weinzierl (ed.), *Musical Instruments in the 21st Century. Identities, Configurations, Practices*, Springer Nature, Singapore 2017, p. 49; D. Hosken, *Music Technology and the Project Studio. Synthesis and Sampling*, Routledge, London/New York 2012, pp. 38–44; Z. Shi, T. Nakano, M. Goto, *Instlistener. An Expressive Parameter Estimation System Imitating Human Performances of Monophonic Musical Instruments*, in: M. Hayes, H. Ko (ed.), *2018 IEEE International Conference on Acoustics, Speech, and Signal Processing. Proceedings*, IEEE Signal Processing Society, Danvers 2018, pp. 581–582.

6 Tanev G., Božinovski A., *Virtual Studio Technology and its Application in Digital Music Production*, in: I. Mishkovski, S. Ristov (ed.), *The 10th Conference for Informatics and Information Technology (CIIT 2013)*, Faculty of Computer Science and Engineering, Skopje 2014, pp. 182–186; A.F.B. Laguna, N.M.P. Valdez, R.C.L. Guevara, *MIDI Implementation of a Kulintang Modal Synthesizer Using the VST 2.4 Standard*,

compositions, as well as the ability to use digital workstations for recording, editing, mixing and mastering[7] served as the basis for defining the subject, the research topic and the purpose of the research presented here.

## The subject of research

The subject of the research[8] presented in this paper is the use of computers, along with dedicated software, in the work of a composer-arranger, and in composition teaching for popular music. The main research topic[9] of the paper are the methods and possibilities for using computers and music software by artists in broadly understood creative work and teaching. The purpose of the research[10] is to determine whether new technologies aid and stimulate popular music composers in their creative work; whether they are used in practice; whether they are useful and become employed in teaching; whether they increase the attractiveness of works produced with hybrid methods; for what other purposes not mentioned in the questionnaire do musicians use computers and music software; and, finally, are digital technologies, when introduced into the creative process, a liability or an opportunity for further creative development?

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in: J.B. Cruz (ed.), *TENCON 2012 IEEE Region 10 Conference*, Institute of Electrical and Electronics Engineers, Inc./IEEE Press, Danvers 2012, pp. 1, 2.

7 D. Etinger, *Tools of the Trade. Digital Audio Workstation Usage Antecedents*, "Informatologia" 2016, 49(1–2), pp. 61, 62; C. Leider, *Digital Audio Workstation*, McGraw-Hill, New York 2004, p. 48; S. Langford, *Digital Audio Workstation. Correcting and Enhancing Audio in Pro Tools, Logic Pro, Cubase, and Studio One*, Focal Press, Burlington 2014, pp. 9–13.

8 J. Sztumski, *Wstęp do metod i technik badań społecznych*, PWN, Warszawa 1984, p. 20; L. Sołoma, *Metody i techniki badań socjologicznych. Wybrane zagadnienia*, Wydawnictwo Uniwersytetu Warmińsko-Mazurskiego w Olsztynie, Olsztyn 1999, p. 13.

9 J. Sztumski, op. cit., p. 20; T. Pilch, T. Baumann, *Zasady badań pedagogicznych. Strategie ilościowe i jakościowe*, Wydawnictwo Akademickie Żak, Warszawa 2001, p. 43; S. Palka, *Metodologia badania. Praktyka pedagogiczna,* Gdańskie Wydawnictwo Psychologiczne, Gdańsk 2006, p. 12; J. Pieter, *Ogólna metodologia pracy naukowej*, Zakład Narodowy im. Ossolińskich-PAN, Wrocław 1967, p. 67; M. Łobocki, *Metody i techniki badań pedagogicznych*, Oficyna Wydawnicza Impuls, Kraków 2003, p. 21.

10 W. Dutkiewicz, *Podstawy metodologii badań do pracy magisterskiej i licencjackiej z pedagogiki*, Wydawnictwo Stachurski, Kielce 2000, p. 50; T. Pilch, *Zasady badań pedagogicznych*, Wydawnictwo Akademickie Żak, Warszawa 1998, p. 8.

## Research method

The present paper utilises the research method[11] of a diagnostic survey[12]. The questionnaire method has been selected from the many existing survey techniques, since it allows for extended and substantial responses on the part of the respondents[13]. The questionnaire was composed of 17 questions, which included open-ended questions, closed-ended questions and hybrid questions, meaning closed-ended questions with the option of entering the respondent's own remarks and suggestions, allowing them to share their perspective on the particular issue.

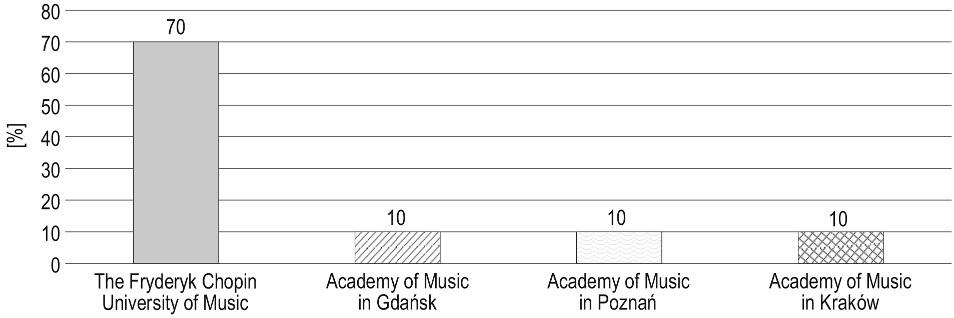


Fig. 1. Percentages of respondents to the questionnaire by institution of employment
*Source*: own research

The questionnaire was administered from 18 January to 5 February 2018 in four Polish academic institutions: Fryderyk Chopin University of Music in Warsaw, Stanisław Moniuszko Academy of Music in Gdańsk, Ignacy Jan Paderewski Academy of Music in Poznań, and the Academy of Music in Kraków.

## Cohort characteristics

The cohort for the survey consisted of 31 people aged 32–75 (22 men and 9 women). The participants held at least a Master of Fine Arts professional degree or higher degrees and titles (including PhD in Fine Arts, Doctor habilitatus,

11 S. Nowak, *Metodologia badań socjologicznych*, PWN, Warszawa 1970, p. 237; M. Łobocki, *Metody badań pedagogicznych*, PWN, Warszawa 1982, p. 56.

12 T. Pilch, op. cit., p. 51.

13 K. Konarzewski, *Jak uprawiać badania oświatowe*, WSiP, Warszawa 2000, p. 138; T. Pilch, op. cit., pp. 86, 87.

and professor). All respondents were active artists and academic lecturers in artistic institutions of higher learning, or retired lecturers. The selection for the cohort was random, since there were no criteria for selecting the respondents or eliminating them once enrolled.

## Survey results

Responses to the questionnaire regarding the use of a computer and music software in the process of composition and arrangement showed that as many as 87% of composers utilised used such technologies for that purpose (cf. Fig. 2). Music software allowed artists to listen to their compositions as they worked on them and to correct harmony or melody errors they detected. The musicians

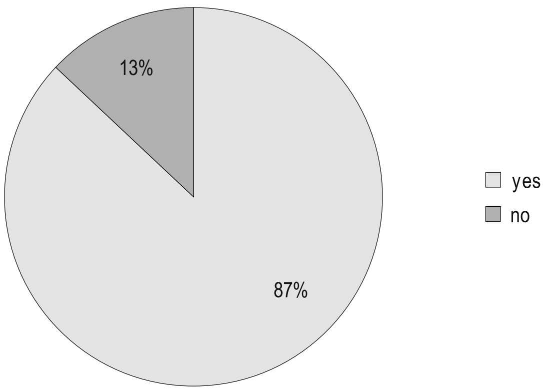


Fig. 2. Do you use a computer with music software in the process of composition and music arrangement?
*Source*: own research

also made use of many sound generators offered by the software and the ability to mix the generated sounds to achieve original effects in their creations. The respondents indicated that music software enables them to work faster than with traditional pen-and-paper notation, which is especially relevant with the current demand for short-term composition over the Internet.

Figure 3 shows that 71% of composers use virtual instruments in their work. The data suggests that the transformation into an information society is happening before our eyes, also impacting the creative work of composers and their output.

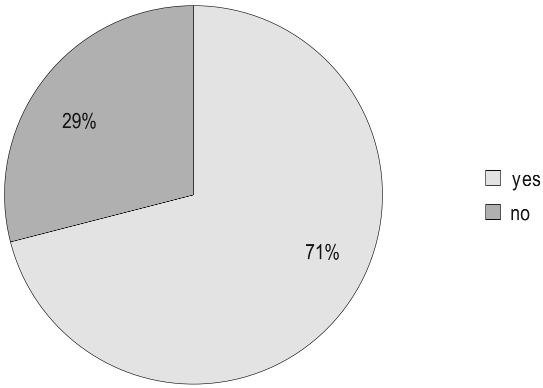


Fig. 3. Do you use virtual instruments? If so, specify the product you use
*Source*: own research

Figure 4 presents the percentages of responses regarding the use of virtual instruments in creative work. Apart from the software specified in the figure, musicians indicated that they use many other virtual instruments, e.g. ethnic, African, Somali and Middle-Eastern. Some respondents used virtual instrumentation software only to replay their students' work, submitted as MIDI files.

Digital instruments are becoming increasingly favoured by popular and electro-acoustic music composers, since they offer a way for them to find new and innovative sounds. Such an effect can be achieved mainly by utilising digital

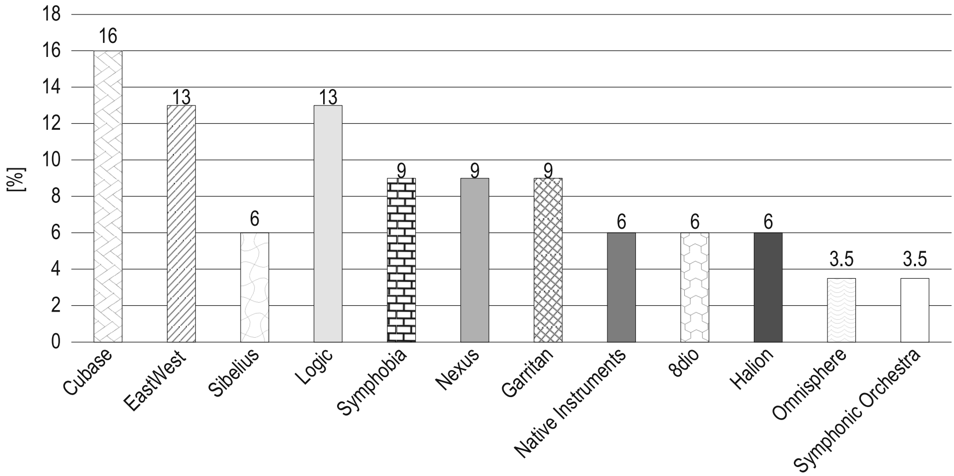


Fig. 4. Virtual instruments used by composers for composition and arrangement
*Source*: own research

sound simulation while processing sound with a computer and computerised devices. Virtual instruments may also be used by film music creators to supplement the orchestra in terms of sonorous effects for the cinematographic musical score, or, in the case of science fiction films, to create voices for fictional beings.

Computer games are currently the main area of development, as well as inspiration, for electronic and electroacoustic music composers. Artists create music for games which corresponds with the given game's story, generate various ambient sounds of the virtual environment and of the moving characters and creatures animated by the programmers.

Answers visualised in Figure 5 indicate that 71% of respondents believe virtual instruments to be a suitable replacement for acoustic ones, provided that they are employed skilfully. Respondents considered virtual instruments essential in education, composition and arrangement, as an aid in preserving the composer's ideas. Digital technology makes it possible for a musical piece's sound and harmony to be heard before it is preformed, e.g. by a live orchestra. Respondents agreed that virtual instrument sound can be used in the final product since they largely resemble traditional instruments in timbre, which also allows for supplementation of acoustic sound in music production.

Several times, respondents raised the issue of using virtual instruments imitating acoustic instruments which are difficult to access in Europe. European music uses characteristic instruments which are partly similar, as evidenced also by music history. Bringing in instruments from other continents often proves excessively costly or impossible, since traditional foreign instruments

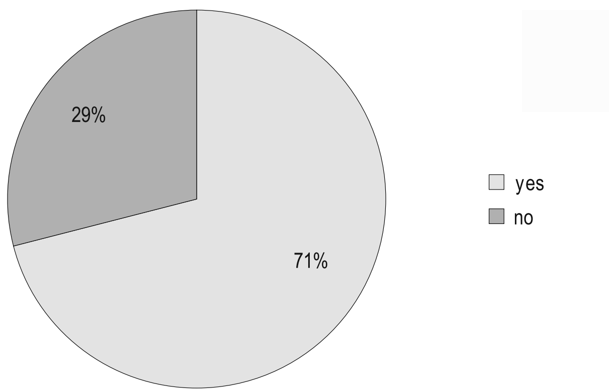


Fig. 5. Do you consider virtual instruments a viable replacement for traditional acoustic instruments? If so, specify how they can be used
*Source*: own research

may be protected as historical artefacts in their countries of origin. Recording an acoustic instrument *in situ* and representing its characteristics as a digital file accessible worldwide makes the timbre of an instrument located, for instance, in Zimbabwe an asset available for music creators everywhere. Digital technology can overcome certain barriers regarding the use of a varied set of instruments, or learning the sound of such geographically distant instruments in the first place, which provides composers with new sound material for their work.

It is worth noting that digital instruments require a new set of skills than a traditionally trained musician, since producing a sound is entirely different from playing acoustic instruments. A person wishing to learn how to play virtual instruments should know that they offer distinct performance possibilities compared to their acoustic counterparts[14].

Figure 6 confirms that the majority of composers (74%) were interested in broadening their knowledge about the practical applications of virtual instrumentation in their work. An analysis of survey results reveals an age limit determining the responses of the participants. Respondents aged 32–50 were interested in broadening their knowledge about digital music technology. Respondents above the age of 50 were not interested in such pursuits, which was also reflected in answers to other questions regarding similar matters.

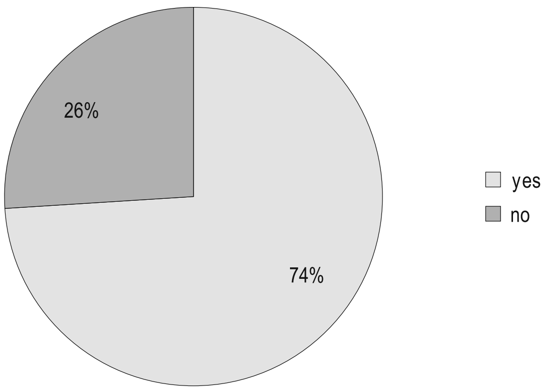


Fig. 6. Are you interested in broadening your knowledge
about the practical application of virtual instruments?
*Source*: own research

[14] A. Pejrolo, R. DeRosa, *Acoustic and MIDI Orchestration for the Contemporary Composer. A Practical Guide to Writing and Sequencing for the Studio Orchestra*, Focal Press, Burlington 2007, pp. 1–26.

The data presented in Figure 7 regards the broadly defined use of music software in composition and arrangement. Responses to the survey reveal that the software suites most commonly used for this purpose are, i.a. Sibelius, Cubase and Finale. Grouping different software suites or using several programmes at once is a deliberate practise by the respondents, since different software may be used for distinct musical purposes.

Figure 8 presents the composers' agreement with the claim that music software improves the creative process (74%). Participants reported that music software's ability to rapidly verify the creative process is especially helpful.

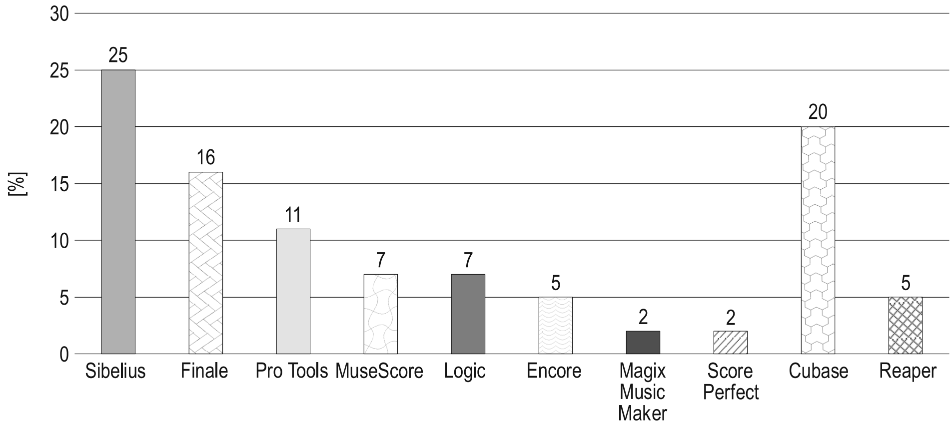


Fig. 7. What music software do you use for composing and arrangement?
*Source*: own research

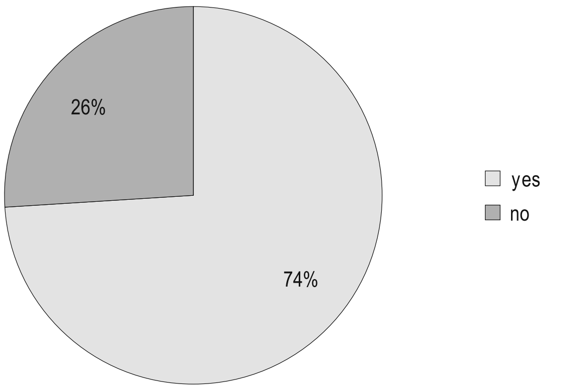


Fig. 8. Do you consider music software an aid to the creative process? If so, in what aspect (please name the software used as such a creative aid)?
*Source*: own research

Sounds of virtual instruments are an inspiration for creating new compositions, arrangements and experimentation with structure and sound. Programmes used for creative inspiration are presented in Figure 9.

The respondents noted in their answers that, on the one hand, composing is made much easier by the ability to listen to the work in progress with the aid of virtual instruments, and the ability to put down notes quickly. The use of a computer with music software may support a musician's artistic

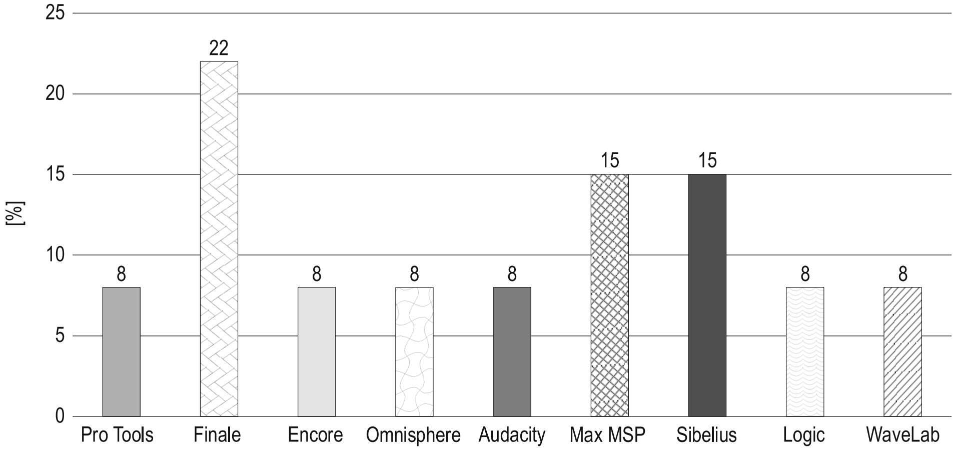


Fig. 9. Software used as aid to the creative process
*Source*: own research

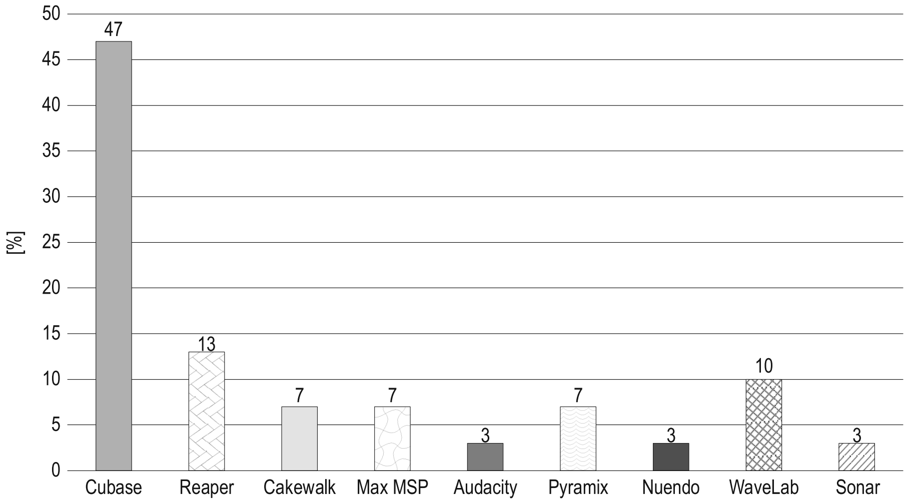


Fig. 10. What music software do you use for music production?
*Source*: own research

development. On the other hand, using such aids in an improper manner – one that replaces the composer's creativity – does not raise the artistic awareness but is an impediment to creative thinking.

Figure 10 presents the music software used by popular music composers in the music production. According to this data, the most popular suites are Cubase, Reaper and Wavelab.

Responses presented in Figure 11 show that most composers (84%), when asked what type of institution of higher learning they work in, named academies of music. Other respondents gave no answer to this question, as well as subsequent questions regarding teaching, which indicates that those respondents were currently uninvolved in teaching students at institutions of higher learning because they were retired lecturers.

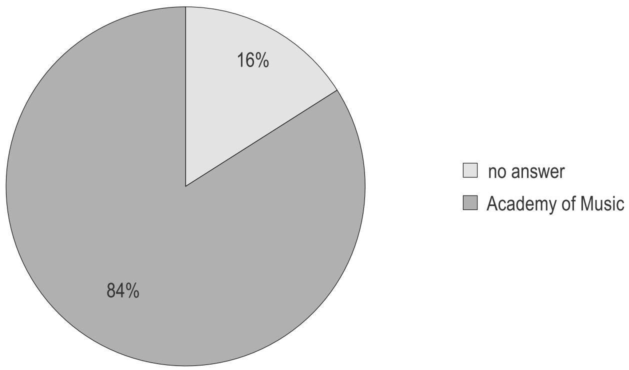


Fig. 11. What type of institution of higher learning do you work in?
*Source*: own research

Figure 12 presents the percentages of questionnaire responses regarding the use of a computer and music software in teaching at an academic institution. 77% of composers answered that they use a computer in instruction. Respondents indicated that using this tool, supplied with the proper music software, can make the classes more attractive and increase the students' interest in the subject matter. Respondents answering in the affirmative were also the ones who were raised in the times when technological development was a constant element of life and society was developing at a fast rate. People answering in the negative or giving no answer were the ones who did not consider a computer as a needed addition to the teaching process. This is due to the fact that in the times of the most rapid technological development those persons already held their diplomas and worked as academic lecturers and thus did not participate in the digital revolution.

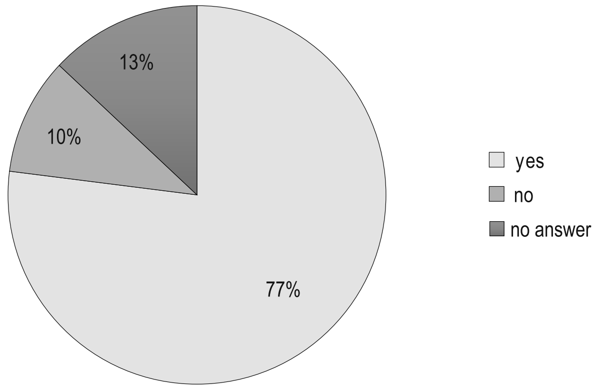


Fig. 12. Do you use a computer with music software in the teaching process?
*Source*: own research

Figure 13 shows how often a computer was used in the teaching process. 39% of the surveyed composers used a computer in every class. 26% used music software often, whereas 6% used it rarely, and 10% very rarely. Respondents who gave no answer to this question (19%) are persons not currently working at a music school (retired), or ones who do not use a computer at all in their teaching work.

The next question posed to composers in the questionnaire was intended to check whether the teaching aids mentioned facilitate faster acquisition of the material by the students. Respondents replied that these devices stimulate the students' creative awareness and their engagement, while the ability to introduce

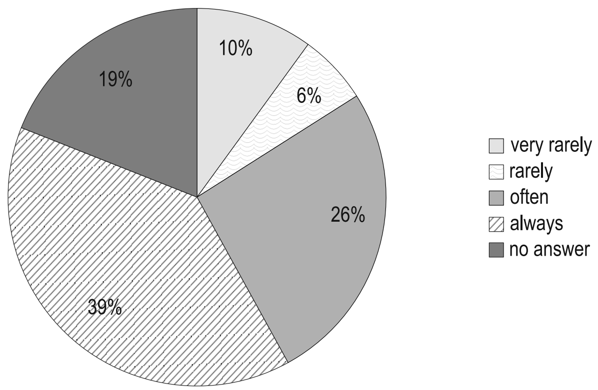


Fig. 13. How often do you use a computer with music software in the teaching process?
*Source*: own research

corrections live and to hear the results facilitates faster acquisition of teaching material. Students have the ability to hear their compositions and arrangements immediately and to improve their skill at writing music scores, which increases their capabilities in expressing and communicating their ideas. Lecturers also pointed out that while analysing a piece of music a computer offers the ability for slower playback, allowing the students to better analyse the hierarchical structure of the composition and to understand it.

In the case of courses such as “electroacoustics or seminar in 21$^{st}$ century music” it is necessary to use a computer in order to present new digital techniques in composition. The computer also serves as a teaching tool in the course “introduction to mixing and mastering”, where it can be used to show the properties of a sound wave in visual form on the screen.

The next question sought to determine which courses at music schools involve music software. Respondents confirmed that computers are used in various courses, such as: “composition, orchestration, composition for film and theatre, electroacoustic composition, arrangement, instrumentation, harmony, counterpoint, introduction to mixing and mastering, seminar in 21$^{st}$ century music, seminar in electroacoustic music, seminar in jazz and popular music, sound engineering, music production, digital recording studio, sound laboratory, digital sound processing, digital sound analysis, musical acoustics, digital technologies in music, sound editing for visual media”.

Figure 14 shows the software used by composers to digitise music notation. The main software packages mentioned by respondents are i.a. Finale, Sibelius, MuseScore.

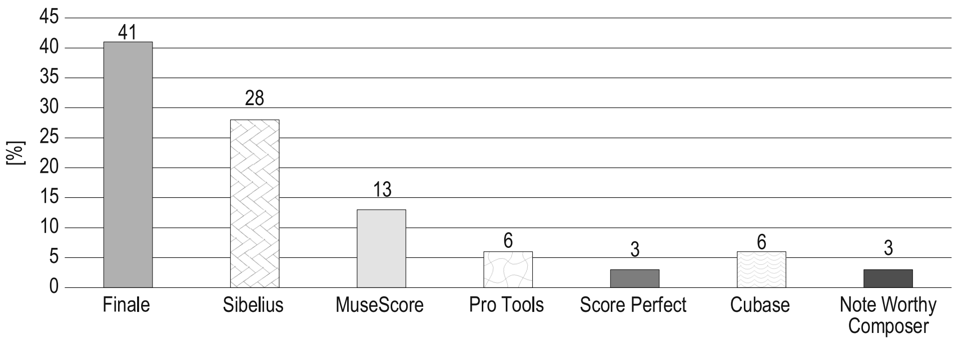


Fig. 14. Do you use music editors for digitising music notation?
If so, which editors do you use?
*Source*: own research

The final Figure (15) presents the musical purposes beyond composition for which respondents declared using a computer. Results show that such devices are mostly used for music post-production (21%), adding effects (19%), and mixing and mastering recordings (19% each). Respondents also mentioned other uses for music software, such as: testing and grading student work, generating sound effects, creating samples, creating audio-visual presentations, programming mixing consoles and music software development. It is also worth noting the use of a computer as a music player and as an element of live electronics for music creation in concerts.

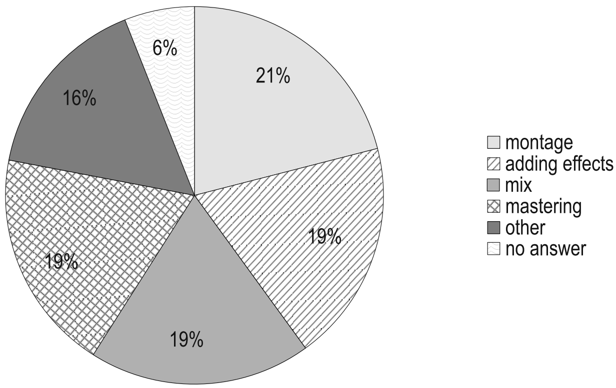


Fig. 15. For what purposes do you use a computer with appropriate music software?
*Source*: own research

## Discussion

Is digital technology needed in music and what purpose should it serve? The need for it is hard to deny in today's technology-dominated world. Does a composer sound surprising when they admit they write their notes manually? Younger artists might ask, why make your life so much harder? Potential audiences are exposed only to the final product and they do not care whether a composer can make traditional notation on a staff. In response to audiences' expectations, composers create ever newer sounds, making their music more innovative, but also more often heeding trends and reception instead of their own compositorial ideas, which is an unfortunate turn in creative development.

An alternative to this dilemma can be found among composers who were formed before computers gained their inextricable hold on everyday life. Most of them have spent long hours imagining the sound of individual instruments or instrumental sections, ruminating, for instance, on the sound of a particular piece played by a philharmonic orchestra in the acoustic environment of a church. Relying on their mind's ear makes a composer receive the final performance as a reflection of their own thoughts. Composing a piece by relying exclusively on the mind's ear is much harder than creating it on the computer, but it also demonstrates a much higher compositorial mastery, which is bolstered by intuition and musical imagination.

Can a computer prove counterproductive in the hands of an inexperienced composer? A less mature musician might be utilising the computer as the only tool for success, which probably indicates that they have not developed musical imagination enough to make music without the aid of a computer. The personal ability to create music is also relevant, since a person who learns composition and arrangement throughout their life develops an independence of thought which is not reliant on any particular software. Probably even an experienced composer will become disaccustomed from proper work with the score if the computer offers them much quicker solutions. Should the number of such poorly trained composers increase, will music not become monotonous, clichéd and derivative from the schema present in the software? Music programmes may be able to impose a schema on the work of such a composer, which, coupled with the fact that many creators use the same software, may lead to factory-like assembly, rather than artistic creation in music.

Experienced composers will not encounter such problems with true creation, since they use the computer only as an expediency, for example, in saving their cores as music notation files so that they look legible in print. Their musical imagination is developed enough that even before writing down their work they know what they mean to achieve and what compositorial techniques they mean to employ to do it. Knowledgeable and sophisticated audiences will be able to verify and discern works which display high compositorial artistry from those created in a mechanical way enabled by digital technologies.

## The acoustic space of traditional and virtual instruments

According to various researchers, acoustic space may be defined quite differently depending on the context in which a given acoustic phenomenon is considered, though it is most often associated with the generation of acoustic waves in a particular environment[15]. For the purpose of this paper, acoustic space is construed as a directional characteristic (direction of acoustic wave generation) of an acoustic musical instrument in an external environment.

Each acoustic instrument limits acoustic waves in a thoroughly distinct fashion, as this element is contingent on:

– the material from which the instrument was made,
– the construction of the instrument itself,
– the mode in which the sound is produced,
– space, i.e. the acoustic environment in which the instrument is located (the interior and its type, or whether it is an open area)
– the pitch and volume of the sound emitted by the acoustic instrument[16].

Research carried out by J. Meyer clearly demonstrates that the acoustic space generated by traditional musical instruments is indeed diverse and exceedingly variable in terms of frequency and sound intensity[17]. Acoustic waves from a single instrument can propagate in radically different directions depending on whether, for instance, the instrumentalist plays an identical melody an octave higher or lower, which indicates that:

– acoustic instruments create a specific and unique acoustic space around themselves,
– the acoustic space of traditional instruments is never homogeneous.

Contemporary microphone techniques and technologies enabling the creation of virtual acoustics rely on state-of-the-art microphone techniques and methods of reproducing sounds (samples) by means of varied speaker systems[18].

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15 R. Zawadzki, *Percepcja w przestrzeni dźwiękowej – akustyka, psychoakustyka i psychologia*, "Audiofonologia" 2003, vol. XXIV, p. 17; S. Bernat, *Słowo wstępne*, in: S. Bernat (ed.), *Dźwięk w krajobrazie jako przedmiot badań interdyscyplinarnych. Sound in Landscape as a Subject of Interdisciplinary Research*, Polihymnia, Lublin 2008, p. 7.

16 Cf. J. Meyer, *Akustik und musikalische Aufführungspraxis. Leitfaden für Akustiker, Tonmeister, Musiker, Instrumentenbauer und Architekte*, PPVMedien, Issue 6th, ext. edition, Bergkirchen 2015.

17 Ibidem.

18 Cf. B. Katz, *Mastering Audio. The Art and the Science*, second edition, Focal Press, Amsterdam, Boston 2007; Ballou G., *Audio Engineering Explained – Professional Audio Recording*, Focal Press, Burlington 2010; Cf. G.Ballou, *Electroacustic Devices. Microphones and Loudspeakres*, Focal Press, Amsterdam, Boston 2009; Cf. B. Bartlett, J. Bartlett, *Practical Recording Techniques*, 6th Edition, Focal Press/Taylor & Francis, Waltham 2013; Cf. J.C. Whitaker, B.K. Benson, *Standard Handbook of Audio and Radio Engineering*, McGraw-Hill, New York 2002; Cf. J.C. Whitaker, *Master Handbook of Audio Production*, McGraw-Hill, New York 2003.

However, current technology is not sufficiently advanced to simulate the acoustic space generated, for example, by an acoustic violin, piano or symphony orchestra in an environment such as a concert hall. The capacities of virtual instruments today are expanding as digital technology continually penetrates increasingly into the domain of music. In simulations of various interiors, the patterns of sound reverberation in a particular room can indeed be adjusted[19]; however, the reproduction of such sounds takes place via the medium of a loudspeaker system, which in itself is incapable of emitting sound exactly as a traditional instrument does. If one were to equip a concert hall with a supreme-quality PA system and reproduce a pre-recorded instrumental performance, the devices involved (from the standpoint of physics) would still not achieve the directional characteristics of the acoustic space characteristic of acoustic instruments.

It may be worthwhile to note that virtual instruments can offer an innovative sound experience thanks to digital surround reproduction systems designated as 5.1, 7.1, 10.2, 22.2, 5.1.2, 5.1.4, 7.1.2, 7.1.4, 7.1.6 etc. However, it is not an immaculately faithful emulation of the propagation of acoustic waves from traditional instruments but a kind of prosthesis which, thanks to various psychoacoustic procedures affecting the human brain, can yield various types of spatial representation (also by way of acoustic illusions).

Virtual instruments make it possible to go beyond the routine patterns, e.g. by creating new samples on the basis of recorded sounds – of a guitar, for instance – and their modulation[20]; nonetheless, the resonance of the acoustic box is completely different from the vibration of loudspeakers, which already suggests that sound simulation may be virtually perfect, but the very acoustic space that the instrument produces cannot be fully achieved through loudspeaker-mediated simulation. The digital modification and modelling of the various sound parameters of virtual instruments[21] are designed to make them sound

19 J. Huopaniemi, M. Karjalainen, V. Välimäki, T. Huotilainen, *Virtual Instruments in Virtual Rooms – A Real-Time Binaural Room Simulation Environment for Physical Models of Musical Instruments*, in: Proceedings of the International Computer Music Conference (ICMC), 1994, pp. 455–462, https://quod.lib.umich.edu/i/icmc/bbp2372.1994?rgn=full+text [accessed: 10.02.2022].

20 Y. Yun, S.H. Cha, *Designing Virtual Instruments for Computer Music*, "International Journal of Multimedia and Ubiquitous Engineering" 2013, vol. 8, no. 5, pp. 173–178.

21 V. Välimäki, T. Takala, *Virtual Musical Instruments – Natural Sound Using Physical Models,* "Organised Sound" 1996, vol. 1, issue 2, pp. 75–86, https://www.researchgate.net/profile/Vesa-Vaelimaeki/publication/231982630_Virtual_musical_instruments_-_Natural_sound_using_physical_models/links/0fcfd50cc24e756655000000/Virtual-musical-instruments-Natural-sound-using-physical-models.pdf [accessed: 16.02.2022].

as similar to acoustic instruments as possible. Numerous composers, students and contemporary music producers take advantage of digital technologies to generate various sounds to a substantial extent and great effect as well, so that even a trained musician may not be able to distinguish the **sound** of a virtual instrument from its acoustic counterpart. However, such aspects apply to phonography (recording) and widely understood popular music, whereas electroacoustic technology today still fails to generate the **acoustic space of a given traditional instrument** (in the context of wave propagation, e.g. in a concert hall). This is an altogether different aspect that tends to be forgotten, but it is worth noting, not in terms of timbre (simulation quality) but where it concerns **acoustics**, i.e. the manner in which acoustic waves propagate in a particular environment. When developing new computer software for the simulation of sounds originating with virtual instruments, an entirely new parameter should be introduced, i.e. the acoustic space, which, as is the case with traditional instruments, should be variable depending on the frequency and intensity of sound. Sound quality (where it involves timbre) relies thoroughly on the analysis of distinct sound characteristics, which do not possess the coverage and counterpart in the analysis of a properly effected acoustic space.

## Conclusions

The results of the conducted survey clearly show that a computer with music software aids composers in their creative and teaching, since it allows for a more rapid verification of the creative process. Digital tools also inspire greater interest among students in the process of music creation. A computer with the appropriate music software is employed mainly by the creators of popular and electroacoustic music, since those genres often include the sound of electronic instruments.

Nowadays, lecturers at institutions of higher learning offer classes involving the use of music software, or, in the case of certain specialised courses, teaching the use of a particular software suite. Composers taking part in the survey indicated that the presentation of musical material being discussed in class is more readily comprehended by students if this material is presented with appropriate digital tools.

The survey results clearly revealed that composers of popular music use dedicated music software for composing and arrangement. Apart from composition and arrangement, musicians named many other uses of that software. Most common uses of the computer were: mixing and mastering, postproduction and editing of the musical material and adding sound effects to recordings. Modern artists search for new sounds which can be aided by virtual instruments. Such instruments may become an inspiration for creating downright pioneering sound, works and musical arrangements.

Digital technology, like any other technical aid, brings noticeable benefits if used rationally and sparingly. The computer should not be treated as an overriding tool, but only as assistance to a qualified composer and arranger. The main application of digital technologies is streamlining the work process and making it more efficient through the use of specialised tools. These should, with proper use, contribute in the long term to the development of the user, not relieve them of the need for responsible and creative thinking.

It is not computers and technological advancement that shape the current and future condition of composition, arrangement and academic teaching of music. It is the people, being aware of the drawbacks and benefits of using digital devices, who must learn to use that technology in moderation. Computer technology, composition, arrangement and academic teaching make constant advancements, which is why we must be prepared for traditional teaching methods to be supported by modern technologies, whose purpose is to improve teaching and to stimulate composers and arrangers of popular music to develop their creative work.

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**Abstract**

Technology, which constantly influences society, is also significant in the work of the composers and arrangers of popular music. This work increasingly involves computers along with dedicated music software. The modern composer is also a music producer and sound engineer, using virtual instruments, music notation editors and digital workstations for music production. This article presents original studies conducted with the participation of Polish composers. The results clearly indicate that contemporary music creation is aided by digital technologies. The application of the computer and computerised devices allows for faster creation of the finished musical product, but can also be a liability to the composer when it becomes an overriding tool, replacing responsibility and creative thinking.

**Key words:** music, composer, digital art, composition, digital technologies in music.